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Stacked 3D memory architecture for power optimization

Abstract

A headset includes a camera a 3D stacked memory configured to store image data captured by the camera, and a System-on-Chip (SoC) configured to process the image data stored in the 3D stacked memory. The 3D stacked memory includes a plurality of first drivers/receivers and a plurality of memory banks that are accessible in parallel. Each memory bank is accessible via a corresponding first driver/receiver. The SoC includes a memory controller with a plurality of second drivers/receivers. The plurality of the second drivers/receivers of the SoC are respectively connected to the plurality of the first drivers/receivers of the 3D stacked memory by a plurality of channels. The SoC and the 3D stacked memory are vertically stacked together. The plurality of the memory banks include at least eight memory banks.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure generally relates to an artificial reality device including a three-dimensional (3D) stacked memory, and in particular relates to using the 3D stacked memory to reduce power consumption of an artificial reality device.

BACKGROUND

(2) Artificial reality is a form of reality that has been adjusted in some manner before presentation

to a user, which may include, e.g., a virtual reality (VR), an augmented reality (AR), a mixed reality (MR), a hybrid reality, or some combination and/or derivatives thereof. An artificial reality system that provides the artificial reality content may be implemented on various platforms, including a head-mounted display (HMD) connected to a host computer system, a standalone HMD, a mobile device or computing system, or any other hardware platform capable of providing artificial reality content to one or more viewers. 3D integrated circuit manufacturing technology may be developed with two or more dice stacked vertically in a 3D structure chip architecture with high storage capacity. Stacking of multiple memory chips increases chip density, provides a reduction in overall package, and improves electrical performance. A 3D stacked dynamic random access memory (DRAM) may be used with processors or memory controllers to implement applications on virtual reality (VR) and augmented reality (AR) devices. AR/VR devices require less power consumption and smaller chip size to enable high-resolution and long duration image capture on significantly power constrained wearable devices. An artificial reality device with a 3D stacked memory requires less power consumption for signal transmission with low data latency.

SUMMARY OF PARTICULAR EMBODIMENTS

(3) Embodiments of this invention may include or be implemented in conjunction with an artificial reality device (e.g., a headset) with a 3D stacked memory. Embodiments of the invention may provide solutions to reduce energy consumption using a customized 3D die-stacking mechanism. Embodiments of the invention may include a customized 3D stacked memory with a memory page configuration and a low power DRAM die, a customized Network-on-Chip (NoC) architecture, a customized memory controller, or a combination of two or more of these.

(4) In particular embodiments, artificial reality devices and methods are described for using 3D stacked memory to reduce power consumption of an artificial reality device such as a headset. The headset comprises a camera, a 3D stacked memory configured to store image data captured by the camera, and a System-on-Chip (SoC) configured to process the image data stored in the 3D stacked memory. The 3D stacked memory comprises a plurality of first drivers/receivers and a plurality of memory banks that are accessible in parallel. Each memory bank is accessible via a corresponding first interconnect. The SoC comprises a memory controller with a plurality of second drivers/receivers. The plurality of the second drivers/receivers are respectively connected to the plurality of the first drivers/receivers of the 3D stacked memory by a plurality of channels. In particular embodiments, the SoC and the 3D stacked memory are vertically stacked together. The plurality of the memory banks each has a page size of 512 bytes or less. The plurality of the memory banks include at least eight memory banks. The plurality of the channels are controlled by using unidirectional and/or bidirectional links. Each channel may operate at 500 MHz or less. In particular embodiments, the memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory. The 3D stacked memory does not use a Double Data Rate (DDR) interface and it does not require to perform impedance matching when transferring stream data from the plurality of memory banks to the SoC.

(5) In particular embodiments, artificial reality devices and methods are described for using a network-on-chip (NoC) architecture to handle multi-channel 3D stacked memory of an artificial reality device such as a headset. In particular embodiments, the headset comprises a 3D stacked memory and a System-on-Chip (SoC). The 3D stacked memory comprises a plurality of memory banks that are accessible in parallel. The SoC comprises a plurality of memory controllers and a network-on-chip (NoC). The NoC comprises a plurality of routers and each router is connected to a plurality of channels. Each memory controller is configured to access the 3D stacked memory via a channel. Each memory controller is respectively connected to a cluster of memory banks. The SoC is configured to determine a channel bandwidth capacity of each channel associated with each cluster; determine a first bandwidth demand for a first set of applications of a subsystem from a first channel associated with a first cluster; and determine whether the first bandwidth demand of the first set of the applications is less than a first channel bandwidth capacity. In response to

determining that the first bandwidth demand of the first set of applications of the subsystem is less than the first channel bandwidth capacity, the SoC is configured to partition the first bandwidth demand to one or more memory banks of the first cluster based on a bandwidth demand of each application in the first set of applications and the first channel bandwidth capacity. The SoC is configured to allocate the first set of the applications to the one or more memory banks of the first cluster to maximize a bandwidth usage of at least one memory bank of the first cluster. In particular embodiments, the SoC is further configured to determine an affinity score between an application producer and a user of data associated with the application; partition the first bandwidth demand to one or more memory banks of the first cluster based on a ranking of one or more affinity scores and the bandwidth demand of each application; and allocate the first set of the applications to the one or more memory banks of the first cluster to maximize a bandwidth usage of at least one memory bank in the first cluster.

(6) In particular embodiments, artificial reality devices and methods are described for achieving high efficiency on a 3D stacked memory an artificial reality device such as a headset. The headset comprises a camera, a 3D stacked memory configured to store image data captured by the camera, and a System-on-Chip (SoC). The 3D stacked memory comprises a plurality of memory banks that are accessible in parallel via the plurality of channels. The SoC comprises a plurality of memory controllers and a network-on-chip (NoC). The SoC and the 3D stacked memory are vertically stacked together. In particular embodiments, each cluster comprises four memory banks. Each memory bank has a page size of 512 bytes or less. The NoC comprises a plurality of routers. Each router is connected to the plurality of the channels. Each memory controller is associated with a channel and is connected to a cluster of memory banks. Each memory controller is configured to operate as an out-of-order scheduler to access each respective memory bank via a channel. The out-of-order scheduler is configured to generate a schedule with an out-of-order sequence of read commands or write commands to control operations of a set of memory banks in each cluster. In particular embodiments, the memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory. The 3D stacked memory does not use a Double Data Rate (DDR) interface and it does not require to perform impedance matching when transferring stream data from the plurality of the memory banks to the SoC. In particular embodiments, each out-of-order scheduler may be further configured to select a memory bank based on a priority of a transaction type associated with each corresponding transaction through the channel; prioritize a set of pages of the memory bank associated with the transaction having a higher priority; and schedule an operation to open the set of pages of the memory bank. In one embodiment, each out-of-order scheduler may be further configured to select a memory bank based on page status, such as open or closed status. In one embodiment, each out-of-order scheduler may be further configured to select a memory bank based on a request associated with a page of a memory bank. In one embodiment, each out-of-order scheduler may be further configured to select a memory bank based on a data transfer direction.

(7) Embodiments of the invention may include or be implemented in conjunction with an artificial reality system. Artificial reality is a form of reality that has been adjusted in some manner before presentation to a user, which may include, e.g., a virtual reality (VR), an augmented reality (AR), a mixed reality (MR), a hybrid reality, or some combination and/or derivatives thereof. Artificial reality content may include completely generated content or generated content combined with captured content (e.g., real-world photographs). The artificial reality content may include video, audio, haptic feedback, or some combination thereof, and any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional effect to the viewer). Additionally, in particular embodiments, artificial reality may be associated with applications, products, accessories, services, or some combination thereof, that are, e.g., used to create content in an artificial reality and/or used in (e.g., perform activities in) an artificial reality. The artificial reality system that provides the artificial reality content may be implemented

on various platforms, including a head-mounted display (HMD) connected to a host computer system, a standalone HMD, a mobile device or computing system, or any other hardware platform capable of providing artificial reality content to one or more viewers.

(8) In particular embodiments, embodiments of the invention may include a variety of subsystems performing methods and functions of artificial reality devices, such as a headset as described herein. The various subsystems may include specialized hardware and integrated circuits and software instructions to facilitate the functions of the headset. In particular embodiments, the subsystems may be functional subsystems operating on one or more processors or memory controllers to execute the software instructions or integrated circuits of the headset. Thus, these are not limited to separate hardware components and software instructions of the headset to implement the solutions as described herein. In particular embodiments, embodiments of the invention may present comprehensive solutions to reduce power consumption in the 3D stacked memory and achieve power efficient data transmission between the 3D stacked memory and the SoC, further improve industrial design of the artificial reality devices. For example, the customized 3D stacked memory may be used in AR applications and devices, computer vision subsystems, or Point of View (PoV) camera subsystems. The customized 3D stacked memory can be used to enable high resolution and long duration captures with limited AR device power budget.

(9) The embodiments disclosed herein are only examples, and the scope of this disclosure is not limited to them. Particular embodiments may include all, some, or none of the components, elements, features, functions, operations, or steps of the embodiments disclosed herein. Embodiments according to the invention are in particular disclosed in the attached claims directed to a device and a method, wherein any feature mentioned in one claim category, e.g., method, can be claimed in another claim category, e.g., system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating a structure of an example headset with a 3D stacked memory and a System-on-Chip (SoC).
- (2) FIG. 2 is a diagram illustrating an example micro-architecture with the 3D stacked memory and the SoC of the headset.
- (3) FIG. 3 illustrates an example method for producing a customized 3D stacked memory of the headset.
- (4) FIG. 4 is a diagram illustrating a Network-on-chip (NoC) topology with the 3D stacked memory of the headset.
- (5) FIG. 5 is a diagram illustrating an example NoC architecture to handle multi-channel traffic of the 3D stacked memory of the headset.
- (6) FIG. 6 illustrates an example method for handling multi-channel traffic of the 3D stacked memory of the headset.

(7) FIG. 7 is a diagram illustrating a traffic profile for each cluster corresponding to the example NoC architecture in FIG. 5 and the method in FIG. 6.

(8) FIG. 8 is a diagram illustrating an example micro-architecture of an out-of-order scheduler used for the 3D stacked memory of the headset.

(9) FIG. 9 is a functional diagram of an example system using an out-of-order scheduler to manage read and write commands for the 3D stacked memory.

(10) FIG. 10 illustrates an example method for using out-of-order schedulers to achieve high efficiency on the 3D stacked memory of the headset.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(11) In particular embodiments, embodiments of the disclosure may include a customized 3D stacked memory with a memory page configuration and a low power DRAM die, a customized NoC architecture, a customized memory controller, or a combination of two or more of these. Artificial reality devices and methods described herein may present a comprehensive solution to reduce power consumption in the 3D stacked memory, achieve power efficient data transmission between a 3D stacked memory and a System-on-Chip (SoC), and further improve industrial design of the artificial reality devices.

(12) In particular embodiments, artificial reality devices and methods may provide a headset which includes a customized 3D stacked memory with a memory page configuration and a low power DRAM die for reducing power consumption of a headset. The headset comprises a camera, a 3D stacked memory configured to store image data captured by the camera, and a System-on-Chip (SoC) configured to process the image data stored in the 3D stacked memory. The 3D stacked memory comprises a plurality of first drivers/receivers and a plurality of memory banks that are accessible in parallel. Each memory bank is accessible via a corresponding first interconnect. The SoC comprises a memory controller with a plurality of second drivers/receivers. The plurality of the second drivers/receivers are respectively connected to the plurality of the first drivers/receivers of the 3D stacked memory by a plurality of channels. In particular embodiments, the SoC and the 3D stacked memory are vertically stacked together. The plurality of the memory banks each has a page size of 512 bytes or less. In particular embodiments, the plurality of the memory banks include at least eight memory banks. The plurality of the channels are controlled by using unidirectional and/or bidirectional links. Each unidirectional and/or bidirectional link may comprise a first driver/receiver, a Die-to-Die (D2D) interconnect, and a second driver/receiver. Each channel may operate at 500 MHz or less. In particular embodiments, the memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory. The 3D stacked memory does not use a Double Data Rate (DDR) interface and it does not require impedance matching when transferring data from the plurality of the memory banks to the SoC.

(13) In particular embodiments, the SoC does not use a dedicated phase-locked loop (PLL) or a delay-locked loop (DLL) for reading data from the memory banks because the SoC operates at the same frequency as the SoC clock. The SoC and the 3D stacked memory operate at different voltages. The memory controller is configured to operate as a scheduler to select and send read commands or write commands to the memory bank. An areal density of the plurality of the number of the channels on the memory die is determined based at least in part on a threshold of a channel capacity and a predefined page size. The plurality of the memory banks is based at least in part on the number of the channels. The number of pages of each memory bank is determined based at least in part on the threshold of a channel capacity and the number of the channels. The number of the plurality of the first drivers/receivers is determined based at least in part on the page size of each memory bank and the areal density of the plurality of the first drivers/receivers on the memory die.

(14) In particular embodiments, a memory die and a SoC die may be vertically stacked together through Die-to-Die (D2D) connections between a plurality of memory banks and the SoC. In particular embodiments, the short D2D interconnects may have a low capacitance value which may

enable the use of low-power and low-voltage input-output drivers. For example, the short D2D interconnects may have a low capacitance value less than 1 pF. The plurality of the channels may be controlled by using unidirectional and/or bidirectional links. Each channel may operate at 500 MHz or less. Stream data may be multiple low-speed parallel stream data. Stream data may be transferred through the short D2D interconnects between the plurality of the memory banks and the SoC. The impedance matching is not needed for the low-speed interface and the short D2D interconnects between the plurality of the memory banks and the SoC.

(15) In particular embodiments, the customized 3D stacked memory may have a small page size of the DRAM memory page of the DRAM on the memory die. The memory page configuration and the low power DRAM die may enable a low power consumption of the 3D stacked memory. The memory die may be a customized low power DRAM die. An areal density of number of the channels on the memory die may be determined based at least in part on a channel capacity threshold and a predefined page size. The plurality of the memory banks may be determined based at least in part on the number of the channels. The number of pages of each memory bank may be determined based at least in part on the channel capacity threshold and the number of the channels. The numbers of the plurality of the first drivers/receivers may be determined based at least in part on the page size of each memory bank and the areal density of the plurality of the first drivers/receivers on the memory die. A memory bank with the smaller page size may be implemented based on an areal density of the plurality of the first drivers/receivers on the memory die.

(16) In particular embodiments, the customized 3D stacked memory may be used to reduce power consumption of an artificial reality device such as a headset. The customized 3D stacked memory with a memory page configuration may provide technical advantages to reduce page size of each memory bank. The customized 3D stacked memory may further lower the power consumption of the 3D stacked DRAM and data transmission between the plurality of the memory banks on the memory die and the SoC on the Soc die.

(17) In particular embodiments, the plurality of the memory banks of the 3D stacked DRAM may each have a small page size of 512 B or less. The 3D stacked DRAM may statistically reduce data transferred through memory arrays of the 3D stacked memory. The 3D stacked DRAM with the small page size may significantly lower the activation power and reduce the number of banks to at least 8 memory banks used in the 3D stacked DRAM.

(18) In particular embodiments, the data transferred between the 3D stacked DRAM and the SoC may be completed through a plurality of the low-speed channels with a 500 MHz bandwidth or less at a lower speed. This significantly reduces data movement inside the memory by using low power interconnects connected to the SoC with an efficient process. The 3D stacked DRAM does not constrain the location or pitch on the location of these connections compared to a memory interface of the WideIO2 architecture.

(19) The low-speed interface does not require a PHY for serialization and deserialization. The data can be sent to the memory using a wide interface (>64 DQ per channel) composed of multiple channels instead of a single channel with low DQ count. The short D2D interconnects have a low capacitance profile (e.g., less than 1 pF) enabling the use of low-power and low-voltage input-output drivers. Impedance matching is not needed in the customized 3D stacked DRAM because of the low-speed interface and short D2D interconnects.

(20) Another advantage of having low-speed interface is that the customized 3D stacked DRAM does not require any additional DLL/PLL inside the SoC to generate the high-speed clocks required for deserialization. These features enable power-efficient interface with the NoC rate matching.

(21) In particular embodiments, artificial reality devices and methods may provide a customized NoC architecture configured to provide solutions to cluster the customized 3D stacked memory to maximize bandwidth and minimize cross cluster communications based on the type of memory and types of applications or services. The artificial reality devices and methods may provide solutions

to dynamically allocate a set of the applications to the one or more memory banks of a cluster to maximize a bandwidth usage of at least one memory bank of the cluster. A headset may include a customized NoC architecture to handle multi-channel 3D stacked memory. The headset comprises a 3D stacked memory and a System-on-Chip (SoC). The 3D stacked memory comprises a plurality of memory banks that are accessible in parallel. The SoC comprises a plurality of memory controllers and a network-on-chip (NoC). The NoC comprises a plurality of routers and each router is connected to a plurality of channels. Each memory controller is associated with a channel and is connected to a cluster of memory banks. The SoC is configured to determine a channel bandwidth capacity of each channel associated with each cluster; determine a first bandwidth demand for a first set of applications from a first channel; determine whether the first bandwidth demand of the first set of the applications is less than a first channel bandwidth capacity of the first channel in a first cluster; in response to determining that the first bandwidth demand of the first set of applications is less than the first channel bandwidth capacity, partition the first bandwidth demand to one or more memory banks of the first cluster based on a bandwidth demand of each application in the first set of applications and the first channel bandwidth capacity; and allocate the first set of the applications to the one or more memory banks of the first cluster to maximize a bandwidth usage of at least one memory bank of the first cluster.

(22) In particular embodiments, the SoC is further configured to determine an affinity score between an application producer and a user of data associated with the application; partition the first bandwidth demand to one or more memory banks of the first cluster based on a ranking of one or more affinity scores and the bandwidth demand of each application; and allocate the first set of the applications to the one or more memory banks of the first cluster to maximize a bandwidth usage of at least one memory bank in the first cluster.

(23) In particular embodiments, the SoC is further configured to identify at least two applications which are initiated by a user associated with the same virtual initiator and transmitted by a router via different channels associated with a cluster; and enable channel interleaving to generate one or more hop transmissions crossing the different channels to allocate the at least two applications to a memory bank of the cluster to maximize a bandwidth usage of the memory bank. In particular embodiments, the SoC is further configured to, in response to determining that the first bandwidth demand of the first set of the applications is not less than the first cluster bandwidth capacity in the first cluster, determine a second bandwidth demand for a second set of applications from a second channel; and partition the second bandwidth demand to allocate at least one application from the first cluster to the second cluster. The at least one application from the first cluster represents a part of an aggregated traffic of a cross-cluster bandwidth. The aggregated traffic of the cross-cluster bandwidth is within a second bandwidth threshold. In particular embodiments, a channel bandwidth capacity associated with each cluster has a first bandwidth threshold.

(24) In particular embodiments, the SoC is further configured to allocate one or more applications to one or more memory banks in the cluster; and select and send a read command or a write command across the set of the memory banks for implementing the one or more applications.

(25) In particular embodiments, the SoC is further configured to partition the first bandwidth demand to one or more memory banks of the first cluster based on the set of the characteristics associated with each memory bank. The set of the characteristics comprises a size, a type, and a locality of the memory bank. The memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory. The SoC and the 3D stacked memory are vertically stacked together. The plurality of the memory banks each has a page size of 512 bytes or less. The plurality of the memory banks include at least eight memory banks. The plurality of the channels are controlled by using unidirectional and/or bidirectional links. Each channel may operate at 500 MHz or less.

(26) In particular embodiments, artificial reality devices and methods may provide a headset which includes a memory page configuration and a low power DRAM with a customized memory

controller for achieving high efficiency on a 3D stacked memory. A headset comprises a camera, a 3D stacked memory configured to store image data captured by the camera, and a System-on-Chip (SoC) configured to process the image data stored in the 3D stacked memory. The 3D stacked memory comprises a plurality of first drivers/receivers and a plurality of memory banks that are accessible in parallel. Each memory bank is accessible via a corresponding first interconnect. The SoC comprises a memory controller with a plurality of second drivers/receivers. The plurality of the second drivers/receivers are respectively connected to the plurality of the first drivers/receivers of the 3D stacked memory by a plurality of channels. In particular embodiments, the SoC and the 3D stacked memory are vertically stacked together. The plurality of the memory banks each has a page size of 512 bytes or less. The plurality of the memory banks include at least eight memory banks. The plurality of the channels are controlled by using unidirectional and/or bidirectional links. Each channel may operate at 500 MHz or less at a lower speed. In particular embodiments, the memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory. The 3D stacked memory does not use a Double Data Rate (DDR) interface and it does not require to perform impedance matching when transferring data from the plurality of the memory banks to the SoC.

(27) In particular embodiments, the SoC does not use a dedicated phase-locked loop (PLL) or a delay-locked loop (DLL) for reading data from the memory banks. The SoC and the 3D stacked memory operate at different voltages. The memory controller is configured to operate as a scheduler to select and send read commands or write commands to the memory bank. An areal density of the plurality of the first drivers/receivers on the memory die is determined based at least in part on a threshold of a channel capacity and a predefined page size. The plurality of the memory banks are based at least in part on the areal density of the plurality of the first drivers/receivers. The number of pages of each memory bank is determined based at least in part on the threshold of a channel capacity and the number of the channels. The number of the plurality of the first drivers/receivers is determined based at least in part on the page size of each memory bank and the areal density of the plurality of the first drivers/receivers on the memory die.

(28) FIG. 1 is a diagram illustrating a structure of an example headset **100** with a 3D stacked memory **110** and a System-on-Chip (SoC) **120**. FIG. 2 is a diagram illustrating an example micro-architecture with the 3D stacked memory and the SoC **120** of the headset **100**.

(29) In particular embodiments, the example headset **100** may include a 3D stacked memory **110**, a SoC **120**, and a camera **150**. The headset **100** is Head-Mounted Display (HMD) that presents content to a user. The headset **100** may include, but not limited to, VR headsets, AR headset, VR glasses, AR glasses, or of any other suitable architecture. The 3D stacked memory **110** may be configured to store image data captured by the camera **150**. As illustrated in FIG. 2, the 3D stacked memory **110** may include a plurality of first drivers/receivers **114** and a plurality of memory banks **112** inside a memory die **111**. The plurality of the second drivers/receivers **124** on the SoC die **119** may be respectively connected to the plurality of the first drivers/receivers **114** on the 3D stacked memory **110** through a plurality of Die-to-Die (D2D) connections **160**. The 3D stacked memory **110** may be accessible in parallel through a plurality of Die-to-Die (D2D) connections **160**. Each memory bank **112** of the 3D stacked memory **110** may be accessible via a corresponding unidirectional and/or bidirectional link **118** (e.g., link **118** in FIG. 2). In particular embodiments, the plurality of the memory banks **112** each may have a plurality of pages **116** (e.g., memory pages). Each page **116** may have a page size of 512 bytes or less. In particular embodiments, the plurality of the memory banks **112** may include at least eight memory banks **112**.

(30) In particular embodiments, the SoC **120** may be configured to process the image data stored in the 3D stacked memory **110**. The SoC **120** may include a Network-on-chip (NoC) **130** and a memory controller **126**. The SoC **120** may be connected to a SoC die **119** with a plurality of second drivers/receivers **124**. The SoC **120** on the SoC die **119** and the 3D stacked memory **110** on the memory die **111** may be vertically stacked together. The plurality of the memory banks **112** may be

accessed by a plurality of channels **134** which are connected to the NoC **130** on the SoC die **119**. The memory die **111** may include multiplexer (Mux) circuitry **117** which is coupled to the plurality of the memory banks **112**. The data movement between die circuitry may happen via unidirectional and/or bidirectional links **118** through the Mux circuitry **117**. As illustrated in FIG. 2, each unidirectional and/or bidirectional link **118** may comprise a first driver/receiver **114**, a D2D interconnect **160**, and a second driver/receiver **124**. The first driver/receiver **114** (e.g., the first driver with receiver) represents a first circuit element which is connected to the memory die **111** and configured to transfer data between the memory die **111** and the SoC die **119**. The second driver/receiver **124** (e.g., the first driver with receiver) represents a second circuit element which is connected to the Soc die **119** and configured to transfer data between the memory die **111** and the SoC die **119**. Each channel **134** may operate at 500 MHz or less. The plurality of unidirectional and/or bidirectional links **118** may be configured to control the plurality of the channels **134** to access the plurality of the memory banks **112**.

(31) In particular embodiments, the headset **100** may include a customized 3D stacked memory with a memory page configuration and a low power DRAM die. In the micro-architecture illustrated in FIG. 2, a memory die **111** and a SoC die **119** may be vertically stacked together through Die-to-Die (D2D) connections **160**. In particular embodiments, the memory controller **126** of the SoC **120** does not use a physical interface (PHY) circuitry to access the 3D stacked memory **110**. The 3D stacked memory **110** does not use a Double Data Rate (DDR) interface to perform impedance matching and it does not require when transferring stream data **115** from the plurality of the memory banks **112** to the SoC **120**. In particular embodiments, the short D2D interconnects **160** may have a low capacitance profile which may enable the use of low-power and low-voltage links **118**. The plurality of the channels **134** may be controlled by using unidirectional and/or bidirectional links **118**. Each channel **134** may operate at 500 MHz or less at a lower speed. In particular embodiments, the short D2D interconnects **160** may have a low capacitance value less than 1 pF. Stream data **115** may be multiple low-speed parallel stream data and be transferred through the short D2D interconnects **160** between the plurality of the memory banks **112** and the SoC **120**. The impedance matching is not needed for the low-speed interface and the short D2D interconnects **160** between the memory die **111** and the SoC die **119**.

(32) In particular embodiments, the SoC **120** does not use a dedicated phase-locked loop (PLL) or a delay-locked loop (DLL) for reading data from the memory banks **112**. In particular embodiments, the SoC **120** chip and the 3D stacked memory **110** operate at different voltages. In particular embodiments, the memory controller **126** is configured to operate as a scheduler to select and send read commands or write commands to the memory bank.

(33) In particular embodiments, the memory die **111** may be a customized low power DRAM die. An areal density of the plurality of the first drivers/receivers **114** on the memory die **111** may be determined based at least in part on a channel capacity threshold and a predefined page size. The plurality of the memory banks **112** may be determined based at least in part on the areal density of the plurality of the first drivers/receivers **114**. The number of pages of each memory bank may be determined based at least in part on the channel capacity threshold and the number of the channels **134**. The number of the plurality of the first drivers/receivers **114** may be determined based at least in part on the page size of each memory bank and the areal density of the plurality of the first drivers/receivers **114** on the memory die **111**. The customized 3D stacked memory may have a small page size of the DRAM memory page of the DRAM on the memory die **111**. A memory bank **112** with the smaller page size may be implemented based on an areal density of the plurality of the first drivers/receivers **114** on the memory die **111**. The plurality of the memory banks **112** of the 3D stacked DRAM may each have a small page size of 512 B or less.

(34) FIG. 3 illustrates an example method **300** for producing a customized 3D stacked memory of the headset **100**. In particular embodiments, the steps **302-314** may be implemented to provide a customized 3D stacked memory page configuration and a low power DRAM die.

(35) At step **302**, a 3D stacked memory **110** may be provided and connected to the memory die **111**. The 3D stacked memory **110** may be configured to store image data captured by a camera **150** of the headset **100**. The 3D stacked memory **110** may include a plurality of first drivers/receivers **114** and a plurality of memory banks **112**. The plurality of the memory banks **112** may be accessible by a plurality of channels **134** in parallel. The plurality of channels **134** may be connected to the NoC **130** which is connected to the SoC die **119**. Each memory bank **112** may be accessible via a corresponding first driver/receiver **114** connected to the SoC die **119**.

(36) At step **304**, a SoC **120** may be provided and configured to process the image data stored in the 3D stacked memory **110**. The SoC **120** may include a memory controller **126** and a plurality of second drivers/receivers **124** which are connected to SoC die **119**. The plurality of the second drivers/receivers **124** may be respectively connected to the plurality of the first drivers/receivers **114** of the 3D stacked memory **110**. The 3D stacked memory **110** may be accessed by the plurality of the channels **134** controlled by the plurality of unidirectional and/or bidirectional links **118**.

(37) In particular embodiments, a customized 3D stacked memory may be implemented with a memory page configuration and a low power DRAM die. The headset may further provide a customized 3D stacked memory with a memory page configuration. The 3D stacked memory **110** may use a 3D stacked DRAM. For example, the channel may have a low channel capacity threshold, such as 8 MB-16 MB. Given the channel capacity threshold, the memory page configuration may reduce a page size of each memory bank **112** to 512 B. The headset **100** may provide an efficient interface associated with the NoC **130** of the SoC **120** with a lower channel capacity.

(38) At step **306**, an areal density of the plurality of the number of the channels **134** on the memory die **111** may be determined based at least in part on a channel capacity threshold and a predefined page size.

(39) At step **308**, the plurality of the memory banks **112** may be determined based at least in part on the areal density of the number of the channels **134** on the memory die **111**. In particular embodiments, specific numbers of memory banks may be determined based on based at least in part on the areal density of the plurality of the first drivers/receivers **114** on the memory die **111**.

(40) At step **310**, the number of pages of each memory bank may be determined based at least in part on the channel capacity threshold and the number of the channels **134** controlled by a plurality of unidirectional and/or bidirectional links **118**.

(41) At step **312**, the numbers of the plurality of the first drivers/receivers **114** may be determined based at least in part on the page size of each memory bank and the areal density of the plurality of the first drivers/receivers **114** on the memory die **111**.

(42) At step **314**, a size of the memory die **111** may be determined based at least in part on the areal density of the plurality of the first drivers/receivers **114** on the memory die **111** and the numbers of the plurality of the first drivers/receivers **114**.

(43) In particular embodiments, the method **300** may be implemented to determine optimal parameters of a page size, a die size, a number of channels, a number of D2D interconnects **160** between the memory die **111** to the SoC die **119**, and an I/O width. Given a low power consumption, more channels for transferring data may be included between the memory die **111** and the SoC die **119** within a small die area of the memory die **111** of the 3D stacked memory **110**. In particular embodiments, the method **300** may be implemented to determine the optimal parameters of the page size, the die size, the number of channels, and the number of D2D interconnects based on a tradeoff analysis for the related parameters.

(44) In particular embodiments, the plurality of the memory banks **112** of the 3D stacked DRAM may each have a small page size of 512 B or less. The 3D stacked DRAM may statistically reduce the amount of data which may be transferred through memory arrays of the 3D stacked memory **110**. The 3D stacked DRAM with the small page size significantly lowers the activation power and reduces the number of banks to 8 memory banks **112** used in the 3D stacked DRAM.

(45) In particular embodiments, the SoC **120** on the SoC die **119** and the 3D stacked memory **110** the memory die **111** are vertically stacked together through a plurality of short D2D interconnects **160** with a low capacitance value less than 1 pF. The utilization of the short D2D interconnects **160** with a low capacitance value also enables the use of low-power and low-voltage input-output drivers. In particular embodiments, the plurality of the channels **134** may be controlled by using links unidirectional and/or bidirectional link **118**. Each unidirectional and/or bidirectional link **118** comprises a first driver/receiver **114**, a D2D interconnect **160**, and a second driver/receiver **124**. Each channel **134** may operate at 500 MHz or less. The data transferred from or to the SoC **120** may be completed through plurality of the channels **134** with a 500 MHz bandwidth. The memory page configuration and the low power DRAM die may also enable a low power consumption of the 3D stacked memory and reduce power consumption of the headset **100** in an artificial reality system.

(46) In particular embodiments, different applications or services may run in a cluster **140**. Different types of RAMs may be suitable for different types of application or services. Some applications and services may not allow or require data transaction between different clusters **140** of the 3D stacked memory **110**. Memory partition and allocation may cause the applications or services provided by application producers to interact with a user occur within a cluster **140**.

(47) FIG. **4** is a diagram illustrating a NoC topology **400** with the 3D stacked memory **110** of the headset **100**. FIG. **5** is a diagram illustrating an example NoC architecture **500** to handle multi-channel traffic of the 3D stacked memory **110** of the headset **100**.

(48) In particular embodiments, the example headset **100** may include a 3D stacked memory **110** and a SoC **120**. As illustrated in FIG. **4**, the NoC topology **400** may include a plurality of routers **132**. Each router **132** in the NoC topology **400** may be connected to a plurality of channels **134**. Each channel **134** may be respectively connected to a memory bank **112** of the 3D stacked memory **110** on the memory die **111**. As illustrated in the example NoC architecture **500** in FIG. **5**, each router **132** may be connected to four channels **134**[1]-[4]. Each channel **134** may be respectively connected to a cluster **140** of corresponding memory banks **112**. For example, a channel **134**[1] may be respectively connected to a cluster **140**[1] of corresponding memory banks **112**[0]-[3] and be associated with a memory controller **126**[1]. In another example, a channel **134**[2] may be associated with a memory controller **126**[2] and be connected to a cluster **140**[2] of memory banks **112**[4]-[7].

(49) In particular embodiments, the plurality of the routers **132** in the NoC topology **400** may be organized in a ring topology as illustrated in FIG. **4**. The NoC topology **400** may provide a mechanism to allocate a plurality of applications or services to a channel corresponding to a dedicated cluster **140**. The cluster **140** may include a group of four memory banks **112** of the 3D stacked memory **110**. The NoC topology **400** may provide channel level parallelism in an AR or VR SOC for improved the bandwidth and lower latency of the 3D stacked memory **110**. The NoC topology **400** may allow to meet workload performance in each cluster **140** without interferences from other workloads running on the SOC.

(50) FIG. **6** illustrates an example method **600** for handling multi-channel traffic of the 3D stacked memory **110** of the headset **100**. Based on the NoC topology **400** and corresponding NoC architecture **500**, the method **600** may provide solutions to cluster the customized 3D stacked memory to maximize bandwidth and minimize cross cluster communications based on the type of memory and types of applications or services. The method **600** may provide solutions to dynamically allocate a set of the applications to the one or more memory banks **112** of a cluster **140** to maximize a bandwidth usage of at least one memory bank **112** of the cluster **140**. The dynamic allocation may be implemented based on a configuration of the SoC **120** with various subsystems to provide software instructions executed by a memory controller **126**. The various subsystems may include specialized hardware and integrated circuits and software instructions to facilitate the functions of the headset. In particular embodiments, the subsystems may be functional subsystems

operating on one or more processors or memory controllers to execute the software instructions or integrated circuits to implement the functions described herein. The method **600** may include the same steps **302-304** from FIG. **3** and the continuation steps **602-614**.

(51) Referring back to step **302**, a 3D stacked memory **110** may be provided and connected to the memory die **111**. The 3D stacked memory **110** may include a plurality of memory banks **112** that are accessible by a plurality of channels **134** in parallel. The 3D stacked memory **110** may be configured to store image data captured by a camera **150** of the headset **100**. The plurality of channels **134** may be connected to the NoC **130** which is connected to the SoC die **119**.

(52) Referring back to step **304**, a SoC **120** may be provided and configured to process the image data stored in the 3D stacked memory **110**. The 3D stacked memory **110** may be accessed by the plurality of the channels **134** which are controlled by the plurality of unidirectional and/or bidirectional links **118**. The SoC **120** may include a plurality of memory controllers **126** and a NoC **130**. The NoC **130** may include a plurality of routers **132** each being connected to a plurality of channels **134**. Each channel **134** may be respectively connected to a cluster **140** of memory banks **112**.

(53) At step **602**, the SoC **120** may be configured to determine a channel bandwidth capacity of each channel associated with each cluster **140** as illustrated in the example NoC architecture **500** in FIG. **5**. FIG. **7** is a diagram illustrating a traffic profile for each cluster **140** corresponding to the example NoC architecture **500** in FIG. **5** and the method **600** in FIG. **6**. As shown in FIG. **7**, applications or services may run in a cluster **140**. Each application or service may be associated with a corresponding application producer or a virtual initiator. Each application or service associated with the virtual initiator may correspond to or require a reading bandwidth (Rd BW) value and a writing bandwidth (Wr BW) value. The reading bandwidth (Rd BW) value may be a range of bandwidth values. The writing bandwidth (Wr BW) value may be a range of corresponding bandwidth values. For example, an application of a computer vision (IP1) associated with a virtual initiator running in cluster **1** may require a reading bandwidth (Rd BW) value and a writing bandwidth (Wr BW) value. An example reading bandwidth (Rd BW) demand may be a range of 2000 MB/s to 3000 MB/s. An example writing bandwidth (Wr BW) demand may be a range of 500 MB/s to 1000 MB/s. For example, the total of the reading bandwidth (Rd BW) demand is 7368 MB/s in the cluster **1**. The total of the reading bandwidth (Rd BW) demand is 3293 MB/s in the cluster **1**. The total of the aggregated bandwidth demand in the cluster **1** is 10.66 GB/s which is less than 16 GB/s. Each bandwidth value may be associated with a corresponding type of memory, such as dynamic random access memory (DRAM) and static random-access memory (SRAM). Different types of memories may be suitable for different types of application or services. In particular embodiments, the 3D stacked memory **110** requires that a channel bandwidth capacity associated with each cluster **140** is less than 16 GB/s. In particular embodiments, the SoC **120** may be configured to determine a set of bandwidth values for an application corresponding to different types of memories in a cluster **140**.

(54) At step **604**, the SoC **120** may be configured to determine a first bandwidth demand for a first set of applications of a subsystem from a first channel **134**[1] via a first channel **134**[1] associated with a first cluster **140**[1]. In particular embodiments, at least one application from the first cluster represents a part of an aggregated traffic of a cross-cluster bandwidth. The channel bandwidth capacity associated with each cluster has a first bandwidth threshold, such as 16 GB/s. The aggregated traffic of the cross-cluster bandwidth is required to be within a second bandwidth threshold, such as 8 GB/s. The amount of memory bandwidth required may be dependent on the type of applications or services.

(55) At step **606**, the SoC **120** may be configured to determine whether the first bandwidth demand of the first set of the applications of the subsystem is less than a first channel bandwidth capacity associated with or in a first cluster **140**[1]. In particular embodiments, a channel bandwidth associated with each cluster has a first bandwidth threshold of 16 GB/s. For example, a VR based

application may require at least memory bandwidth of 16 GB/s.

(56) At step **608**, in response to determining that the first bandwidth demand of the first set of the applications of the subsystem is less than a first channel bandwidth capacity of the first channel in the first cluster **140[1]**, the SoC **120** may be configured to partition the first bandwidth demand to one or more memory banks **112** of the first cluster **140[1]** based on a bandwidth demand of each application, the first channel bandwidth capacity, and/or a bandwidth density of the first channel **134[1]**. The bandwidth density may represent a reuse factor of a channel and be defined as one over the number of memory banks in a cluster. Based on the first channel bandwidth capacity and the first channel bandwidth density, the SoC **120** may be configured to determine whether to allocate one or more applications associated with corresponding virtual initiators to the 3D stacked memory **110** at all or to allocate one or more applications to a SRAM or a conventional DRAM. For example, one application may be more suited to use a SRAM or a conventional DRAM.

(57) In particular embodiments, the SoC **120** may be configured to partition the first bandwidth demand to one or more memory banks **112** of the first cluster **140[1]** based on one or more characteristics associated with each memory bank **112**. The set of the characteristics associated with each memory bank may include a size, a type, and a locality of the memory banks **112**. The SoC **120** may be configured to store relationships between corresponding application producers and users of data associated with the applications to ensure that the localities of the memory banks **112** associated with the corresponding applications are maintained in the same cluster **140**.

(58) At step **610**, the SoC **120** may be configured to allocate the first set of the applications to the one or more memory banks **112** of the first cluster **140[1]**. In particular embodiments, the SoC **120** may be configured to allocate the first set of the applications to the one or more memory banks **112** of the first cluster **140[1]** based on the types of applications or services to maximize a bandwidth usage of at least one memory bank **112** of the first cluster **140[1]**. In particular embodiments, the applications or services within each cluster **140** may be selected based on bandwidth requirements of the applications and how the application producer and a user of data associated with the application are related to each other. For example, the SoC **120** may be configured to determine an affinity score between an application producer and a user of data associated with the application associated with a cluster **140**. The one or more memory banks **112** may store the affinity score representing relationships between corresponding application producers and users of data associated with the first set of the applications. The user associated with the headset **100** may initiate the application. The SoC **120** may be configured to partition the first bandwidth demand to one or more memory banks **112** of the first cluster **140[1]** based on a ranking of one or more affinity scores associated with the applications and the bandwidth demand of each application in the cluster **140[1]**. Further, the SoC **120** may be configured to allocate the first set of the applications to the one or more memory banks **112** of the first cluster **140[1]** to maximize a bandwidth usage of at least one memory bank **112** of the first cluster **140[1]**.

(59) In particular embodiments, the SoC **120** may be configured to identify at least two applications associated with the same virtual initiator. The at least two applications may be transmitted by a router **132** via different channels **134** each associated with a cluster **140**. The SoC **120** may be configured to enable channel interleaving to generate one or more hop transmissions crossing the different channels **134** to allocate the at least two applications to a memory bank **112** of the cluster **140** to maximize a bandwidth usage of the memory bank **112**.

(60) At step **612**, in response to determining that the first bandwidth demand of the first set of the applications of the subsystem is not less than a first channel bandwidth capacity of the first channel in the first cluster **140[1]**, the SoC **120** may be configured to determine a second bandwidth demand for a second set of applications of the subsystem from a second channel **132[2]** associated with a second cluster **140[2]**.

(61) At step **614**, the SoC **120** may be configured to partition the second bandwidth demand to allocate at least one application from the first cluster **140[1]** to the second cluster **140[2]**. Two

related applications or services may be allocated to different clusters **140** of the 3D stacked memory **110** based on the bandwidth values. The 3D stacked memory **110** requires that an aggregated traffic of the cross-cluster bandwidth is within a second bandwidth threshold of 8 GB/s.

(62) In particular embodiments, the SoC **120** may be further configured to allocate one or more applications to one or more memory banks **112** in the clusters **140**. The SoC **120** may be further configured to select and send a read command or a write command across the set of the memory banks **112** for implementing the one or more applications.

(63) The memory partition and allocation implemented in the method **600** may cause an application or related applications provided to a user to occur within a cluster **140**. Therefore, the application or the related applications do not require data transaction between different clusters **140**. Further, VR applications may induce a large memory and memory bandwidth footprint. The memory partition and allocation implemented in the method **600** may reduce memory consumption or power consumption during runtime.

(64) In particular embodiments, artificial reality devices and methods may provide a headset **100** which includes a memory page configuration and a low power DRAM with a customized memory controller **126** for achieving high efficiency on a 3D stacked memory of a headset **100**. In particular embodiments, the memory controller **126** may be configured to be a content-addressable memory (CAM) based an out-of-order scheduler **128**. The out-of-order scheduler **128** may be used to manage incoming read/write commands and selectively issue the corresponding commands to the 3D stacked memory **110**. The operation may optimize the operation efficiency within a page **116** of a memory bank **112**. For example, multiple operations may be performed on data within the same page even though the operations are not received in a sequential order. Further, the out-of-order scheduler **128** may be used with 4 memory banks on the 3D stacked memory **110** to reduce the power-consumption and the size of a scheduler to meet high bandwidth efficiency of the headset **100**.

(65) Referring back to FIG. **1**, a headset **100** may include a camera **150**, a 3D stacked memory **110**, and a SoC **120**. The 3D stacked memory may be configured to store image data captured by the camera **150**. The 3D stacked memory **110** may include a plurality of memory banks **112** that are accessible in parallel via the plurality of channels **134**. The SoC **120** may include a plurality of memory controller **126** and a NoC **130** element. The SoC **120** and the 3D stacked memory **110** may be vertically stacked together. Each cluster **140** may include four memory banks **112**. Each memory bank has a page size of 512 bytes or less.

(66) Referring back to FIGS. **4-5**, the NoC **130** may include a plurality of routers **132**. Each router **132** may be connected to the plurality of the channels **134**. Each memory controller **126** may be connected to a cluster **140** of memory banks **112**. Each memory controller **126** may be configured to operate as an out-of-order scheduler **128** to access each respective memory bank **112** via a channel **134**.

(67) FIG. **8** is a diagram illustrating an example micro-architecture of an out-of-order scheduler **128** used for the 3D stacked memory **110** of the headset **100**. Each router **132** may be connected to four channels **134**. Each out-of-order scheduler **128** may be associated with a channel **134** and may be respectively connected to a cluster **140** of memory banks **112**. For example, a channel **134** may be associated with the out-of-order schedulers **128[0]-[3]**. In the cluster **140[1]**, the set of out-of-order schedulers **128[0]-[3]** may be connected to respective memory banks **112[0]-[3]**. Similarly, In the cluster **140[2]**, the set of out-of-order schedulers **128[4]-[7]** may be connected to respective memory banks **112[4]-[7]**. Each memory bank **112** may be a part of a customized 3D stacked memory **110** with a memory page configuration and a low power DRAM die as described above. The out-of-order scheduler **128** may be configured to generate a schedule with an out-of-order sequence of read commands and write commands to control operations and memory traffic of a set of memory banks **112** in each cluster **140**.

(68) FIG. **9** is a functional diagram of an example system **900** using an out-of-order scheduler **128**

to manage read or write commands for the 3D stacked memory **110**. In particular embodiments, the system **900** may include an example out-of-order scheduler **128** connected to a set of memory banks **112**. The out-of-order scheduler **128** is a content-addressable memory (CAM) based scheduler and includes a set of functional circuit elements. The set of the functional circuit elements may include read commands interfaces **902[1]-[n]**, write commands interfaces **904[1]-[n]**, read data interfaces **906[1]-[n]**, a read CAM **908**, a write CAM **910**, a scheduler with a command (CMD) interface **912**, and read data interfaces **914[1]-[n]**.

(69) The write CAM **910** may be configured to operate as a write staging buffer to store incoming write commands inside the out-of-order scheduler **128** in response to write requests. The write commands may be queued to write data to at least one memory bank **112** of the 3D stacked memory **110** through a write commands interface **904**. The scheduler **128** may be configured to maintain open transactions per open page. The out-of-order scheduler **128** may be configured to maintain memory bank status to indicate which memory banks are open or closed. The read CAM **908** may be configured to operate as a read staging buffer to store incoming read commands for reading data from at least one memory bank **112** of the 3D stacked memory **110** through the read commands interface **902** in response to read requests. The out-of-order scheduler **128** may be configured to indicate whether the scheduler **128** issues write commands or read commands to a memory bank **112**. The larger the memory banks, the larger the read command queues or write command queues will be. The read command queue for processing read requests is separate from the write command queue for processing write requests. The scheduler **128** may be configured to prioritize read commands and provide read command queue with a higher bandwidth compared to a write command queue. For example, the out-of-order scheduler **128** may prioritize instructions for page hits, support urgent requests, and minimize read-write switches. The system **900** may include an out-of-order scheduler **128** with a command interface **912** to schedule a read command or a write command to at least one memory bank **112** of the 3D stacked memory **110**. The out-of-order scheduler **128** may read data from the memory bank **112** through read data interfaces **914[1]-[n]** from the memory banks **112**. The system **900** may include read data interfaces **906[1]-[n]** for reading data from the read data interfaces **914[1]-[n]**.

(70) FIG. **10** illustrates an example method **1000** for using out-of-order schedulers **128** to achieve high efficiency by reducing power consumption and a memory size of the 3D stacked memory **110** of the headset **100**. The method **1000** may include the same operations **302-304** from FIG. **3** and the continuation steps **1002-1010**.

(71) Referring back to step **302**, a 3D stacked memory **110** may be provided and connected to the memory die **111**. The 3D stacked memory **110** may include a plurality of memory banks **112** that are accessible by a plurality of channels **134** in parallel. The 3D stacked memory **110** may be configured to store image data captured by a camera **150** of the headset **100**. The plurality of channels **134** may be connected to the NoC **130** which is connected to the SoC die **119**.

(72) Referring back to step **304**, a SoC **120** may be provided on a SoC die **119** and configured to process the image data stored in the 3D stacked memory **110**. The 3D stacked memory **110** may be accessed by the plurality of the channels **134** controlled by the plurality of unidirectional and/or bidirectional links **118**. The SoC **120** may include a plurality of memory controllers **126** and a NoC **130**. The NoC **130** may include a plurality of routers **132** each being connected to a plurality of channels **134**. Each channel **134** may be respectively connected to a cluster **140** of memory banks **112**. Each memory controller **126** may be associated with a channel **134** and be connected to a cluster **140** of memory banks **112**.

(73) At step **1002**, each memory controller **126** may be configured to operate as an out-of-order scheduler **128** to access each respective memory bank **112**. As illustrated in FIG. **9**, an out-of-order scheduler **128** may be used to manage and selectively issue incoming read and write commands to one or more memory banks **112** of the 3D stacked memory **110**. In particular embodiments, multiple operations on data within the same page **116** of a memory bank **112** may be performed

together even though those operations are not received in sequential order.

(74) At step **1004**, the out-of-order scheduler **128** may be configured to generate a schedule with an out-of-order sequence of read commands or write commands to control operations of a set of memory banks **112** in each cluster **140**. For example, the out-of-order scheduler **128** may selectively issue either read commands or write commands to the memory banks **112** to optimize the operation efficiency within a page **116** of a memory bank **112** while reducing the power-consumption. In particular embodiments, each out-of-order scheduler **128** may be configured to determine a priority to select a command. Commands may be selected based on page status.

(75) In particular embodiments, commands to open some pages may be sent out ahead of commands to close other pages. Further, commands may be selected to improve efficiency. In particular embodiments, out-of-order scheduler **128** may be configured to set a read command priority of opening a memory bank **112** over another memory bank **112**. For example, the out-of-order scheduler **128** may determine a priority of certain data stream over another data stream based on a user selection.

(76) At step **1006**, the out-of-order scheduler **128** may be configured to select a memory bank **112** based on a priority of a transaction type associated with each transaction. The transaction may be associated with an application or service which runs in a cluster **140**. The application or service is initiated by an application producer to a user associated with the headset **100**.

(77) At step **1008**, the out-of-order scheduler **128** may be configured to prioritize the pages **116** of the memory bank **112** associated with the higher priority transaction. The out-of-order scheduler **128** may be configured to determine a set of pages **116** of the memory bank **112** associated with the transaction having a higher priority. For example, selecting a memory bank **112** to send the command may be based on a higher priority transaction associated with the memory bank **112**. In one embodiment, each out-of-order scheduler **128** may be further configured to select a memory bank **112** based on a request associated with a page **116** of the memory bank **112**. In one embodiment, each out-of-order scheduler **128** may be further configured to select a memory bank **112** based on a data transfer direction to or from the memory bank **112**. In one embodiment, each out-of-order scheduler **128** may be further configured to select a memory bank **112** based on page status, such as open or closed status.

(78) At step **1010**, the out-of-order scheduler **128** may be configured to schedule an operation to open the set of pages **116** of the memory bank **112**. Each out-of-order scheduler **128** may be configured to determine whether there is a critical precharge command based on a timing cycle. In response to determining that there is no critical precharge command, each out-of-order scheduler **128** may be configured to schedule a write command or a read command. Each out-of-order scheduler **128** may further be configured to schedule a precharge command after the write command or the read command is executed. Each out-of-order scheduler **128** may be configured to schedule an activation command after the precharge command is executed.

(79) In particular embodiments, each out-of-order scheduler **128** may be configured to determine a priority to select a command. Commands may be selected based on page status, such as an open status, closing status, or close status. In particular embodiments, commands to open a page **116** may be sent out ahead of commands to close a page **116**. Further, commands may be selected to improve efficiency. In particular embodiments, out-of-order scheduler **128** may be configured to set a priority of a command to open a memory bank over sending a command to read another open bank.

(80) Table 1 shows a traffic analysis of an out-of-order scheduler **128** associated with corresponding memory structure and memory area analysis. As shown in Table 1, the out-of-order scheduler **128** with 4 memory banks **112** in a cluster **140** meets bandwidth requirements for a 3D stacked memory **110**. In particular embodiments, the 3D stacked memory **110** with 4 memory banks and the out-of-order schedulers **128** in the cluster **140** may provide 85% bandwidth efficiency and lower latency of the 3D stacked memory **110**.

(81) TABLE-US-00001 TABLE 1 Band- Average width Effici- Memory Memory Structure Area
ency Latency Baseline 16 channels × 4 banks × 2 MB 15.2 mm.sup.2 60% 239 ns 8 Banks 16
channels × 8 banks × 1 MB 3.2 mm.sup.2 70% 167 ns 4 Banks 16 channels × 4 banks × 2 MB
1.248 mm.sup.2 85% 164 ns with an out-of- order scheduler

(82) In particular embodiments, other advantages of using an out-of-order scheduler **128** may include managing incoming read and write commands and selectively issuing the corresponding commands to the 3D stacked memory **110** to optimize the operation efficiency within a page **116**. The pages **116** may be configured to be open before the data is written to the pages **116**. In particular embodiments, the process may increase the efficiency of data transmission of the data bus. Further, using out-of-order schedulers **128** in a 3D stacked memory **110** may further simplify a 3D stacked memory **110** with 4 memory banks **112** and 4 out-of-order schedulers **128** in a cluster **140**, which decreases the scheduler size with improved power efficiency and lower density and further optimizes operating ranges with a bandwidth restrained within a certain limit.

(83) In particular embodiments, a 3D stacked memory **110** with 4 memory banks **112** in a cluster **140** can provide enough performance for the traffic characteristics. With only 4 memory banks in each cluster **140**, the size of the out-of-order scheduler **128** may be decreased to ¼ of the size of a traditional scheduler. Therefore, the out-of-order scheduler **128** provides an efficient solution with a small size of a memory device.

(84) Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

(85) The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages.

Claims

1. A headset, comprising: a camera; a 3D stacked memory configured to store image data captured by the camera, the 3D stacked memory having a plurality of first drivers/receivers and a plurality of memory banks that are accessible in parallel, wherein each memory bank is accessible via a corresponding first driver/receiver; and a System-on-Chip (SoC) configured to process the image data stored in the 3D stacked memory, the SoC having a memory controller with a plurality of second drivers/receivers, wherein the plurality of the second drivers/receivers are respectively connected to the plurality of the first drivers/receivers of the 3D stacked memory by a plurality of

channels, wherein: the SoC and the 3D stacked memory are vertically stacked together; the plurality of the memory banks each has a page size of 512 bytes or less; the plurality of the memory banks include at least eight memory banks; and the plurality of the channels are controlled by unidirectional and/or bidirectional links.

2. The headset of claim 1, wherein the memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory.

3. The headset of claim 1, wherein the 3D stacked memory does not use a Double Data Rate (DDR) interface and does not require to perform impedance matching when transferring stream data from the plurality of memory banks to the SoC.

4. The headset of claim 1, wherein the SoC does not use a dedicated phase-locked loop (PLL) or a delay-locked loop (DLL) for reading data from the memory banks, and wherein the SoC operates at the same frequency as a SoC clock.

5. The headset of claim 1, wherein the SoC and the 3D stacked memory operate at different voltages, and each channel operates at 500 MHz or less.

6. The headset of claim 1, wherein the memory controller is configured to operate as a scheduler to select and send read commands or write commands to the memory bank.

7. The headset of claim 6, wherein an areal density of the number of the channels on a memory die is determined based at least in part on a threshold of a channel capacity and a predefined page size.

8. The headset of claim 7, wherein the plurality of the memory banks is based at least in part on the areal density of the plurality of the number of the channels.

9. The headset of claim 7, wherein the number of pages of each memory bank is determined based at least in part on a channel capacity threshold and the number of the channels.

10. The headset of claim 9, wherein the numbers of the plurality of the first drivers/receivers are determined based at least in part on the page size of each memory bank and an areal density of the plurality of the first drivers/receivers on a memory die.

11. A method comprising: providing a 3D stacked memory configured to store image data captured by a camera, the 3D stacked memory comprising a plurality of first drivers/receivers and a plurality of memory banks that are accessible by a plurality of channels in parallel, wherein each memory bank is accessible via a corresponding first driver/receiver; and providing a System-on-Chip (SoC) configured to process the image data stored in the 3D stacked memory, the SoC comprising a memory controller with a plurality of second drivers/receivers, wherein the plurality of the second drivers/receivers are respectively connected to the plurality of the first drivers/receivers of the 3D stacked memory by the plurality of the channels, wherein: the SoC and the 3D stacked memory are vertically stacked together; the plurality of memory banks each has a page size of 512 bytes or less; the plurality of memory banks include at least eight memory banks; and the plurality of the channels are controlled by a plurality of unidirectional and/or bidirectional links.

12. The method of claim 11, wherein the memory controller of the SoC does not use a physical interface (PHY) circuitry to access the 3D stacked memory.

13. The method of claim 11, wherein the 3D stacked memory does not use a Double Data Rate (DDR) interface and it does not require to perform impedance matching when transferring stream data from the plurality of memory banks to the SoC.

14. The method of claim 11, wherein the SoC does not use a dedicated phase-locked loop (PLL) or a delay-locked loop (DLL) for reading data from the memory banks.

15. The method of claim 11, wherein the SoC and the 3D stacked memory operate at different voltages, and each channel operates at 500 MHz or less.

16. The method of claim 11, wherein the memory controller is configured to operate as a scheduler to select and send read commands or write commands to the memory bank.

17. The method of claim 16, further comprising determining an areal density of the number of the channels on a memory die based at least in part on a threshold of a channel capacity and a predefined page size.

18. The method of claim 17, further comprising determining the plurality of the memory banks based at least in part on the number of the channels.

19. The method of claim 17, further comprising determining the number of pages of each memory bank based at least in part on a channel capacity threshold and the number of the channels.

20. The method of claim 19, further comprising: determining the numbers of the plurality of the first drivers/receivers based at least in part on the page size of each memory bank and an areal density of the plurality of the first drivers/receivers on a memory die; and determining a size of the memory die based at least in part on the areal density of the plurality of the first drivers/receivers on the memory die and the numbers of the plurality of the first drivers/receivers.
